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A Molecular Orbital Theory of Reactivity in Aromatic Hydrocarbons

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In the search for a quantitative correlation between reactivity and electronic configuration of aromatic hydrocarbons, the electron density, at each carbon atom, of the highest occupied π -orbital in the ground state of the molecule is calculated by means of the LCAO method. Comparing the result of such a calculation on fifteen condensed aromatic hydrocarbons with their chemical reactivities, we find that the position at which the electron density is largest is most readily attacked by electrophilic or oxidizing reagents.

It is, therefore, concluded that distinct from other π -electrons the pair of π -electrons occupying the highest orbital, which is referred to as *frontier* electrons, plays a decisive role in chemical activation of these hydrocarbon molecules. The theoretical significance of this discrimination of the *frontier* electrons in relation to the chemical activation is discussed.

INTRODUCTION

CONSIDERABLE progress has been made of late in the quantum-mechanical treatments for explaining the chemical reactivity of organic compounds. Especially, the problem of predicting the reactivity of an organic molecule in a particular position has been solved to some extent by means of the LCAO molecular orbital method in the case of the well-known "orienting effect" of substituents in aromatic nuclei. On treating such a problem, two methods have been applied by several authors: the one may be called the "transition state method"^{1,2} and the other the " π -electron density method."^{3,4}

In the former method the configuration of transition complex is assumed *a priori* on the basis of some other theoretical or empirical arguments, and this method is usually confined to tedious calculations, as it is necessary to compute the energies of all the possible configurations.*

In the latter method, on the other hand, the π -electron density was computed as a resultant of the contributions of all π -electrons in the molecule, and no distinction between individual π -electrons has ever been considered. The defect of the method arising from this indiscrimination appears most evidently in the treatment of unsubstituted aromatic hydrocarbons, in which the π -electron density has always the same value at each position in the molecule according to the Coulson-Rushbrooke theorem,² so that this method is of no use for predicting the reactivity of different positions in such molecules.†

¹ G. W. Wheland, J. Am. Chem. Soc. **64**, 900 (1942).

² H. C. Longuet-Higgins, J. Chem. Phys. **18**, 283 (1950).

³ G. W. Wheland and L. Pauling, J. Am. Chem. Soc. **57**, 2086 (1935).

⁴ H. C. Longuet-Higgins and C. A. Coulson, Trans. Faraday Soc. **43**, 87 (1947).

* This method has never been applied to unsubstituted hydrocarbons except to naphthalene (see reference 1). Whether such a calculation is successfully applicable to the condensed aromatic hydrocarbons which are treated in the present paper, is not known.

† In relation to this problem two other quantum-mechanical methods have been proposed in which the distribution of π -electron density is discussed without using LCAO method. The one is called by O. Schmidt (Z. physik. Chem. **39**, 59 (1938); **42**, 83 (1939); **44**, 185, 194 (1939)) the "Kastenmodel" method in

In the present paper it is shown that on the ground of the latter method, if we distinguish the pair of π -electrons in the highest occupied orbital in the ground state from the others and assume that this pair of π -electrons plays a decisive role in the reaction in question, we can obtain an illuminating explanation of the difference of reactivity at each position in a molecule.

In the first place we will report on the treatment of unsubstituted aromatic hydrocarbons. The orienting effect of substituents in aromatic nuclei will be treated in the next publication.

CALCULATION OF ELECTRON DENSITY

In the present paper our treatment is restricted to condensed aromatic hydrocarbons which are "alternant."⁵ Namely, they consist of several fused six-membered rings and have the Kékulé-type structure in their ground state.

The energy spectrum can be obtained by the LCAO method, using one-electron approximation function

$$\Psi = \sum_{i=1}^{2N} c_i \varphi_i, \quad (1)$$

where φ_i is the $2p\pi$ -atomic orbital of the i th carbon atom.‡

The secular equations for a series of aromatic hydrocarbons are reduced according to the modes shown in Appendix, where the symmetry operations σ_h , σ_v , and i are not considered.

The density of each electron occupying the n th electronic level at the i th carbon atom is represented by $|c_i|^2$, where c_i is the coefficient in Eq. (1) corresponding to the n th eigenvalue.

which certain ambiguities are involved as to the model used, resulting in an insufficient accuracy of the method. The other is the "molecular diagram" method after A. and B. Pullman (Rev. sci. **84**, 145 (1946); **86**, 219 (1948); Chem. Abst. **43**, 6905 (1949)) which is almost inapplicable to complex molecules, owing to the laboriousness in calculating the contributions of all possible bond structures, including excited ones, to the normal state of the molecule.

⁵ C. A. Coulson and H. C. Longuet-Higgins, Proc. Roy. Soc. (London) **A192**, 16 (1947).

‡ All σ -electrons which form a closed shell in the ground state are left out of account.

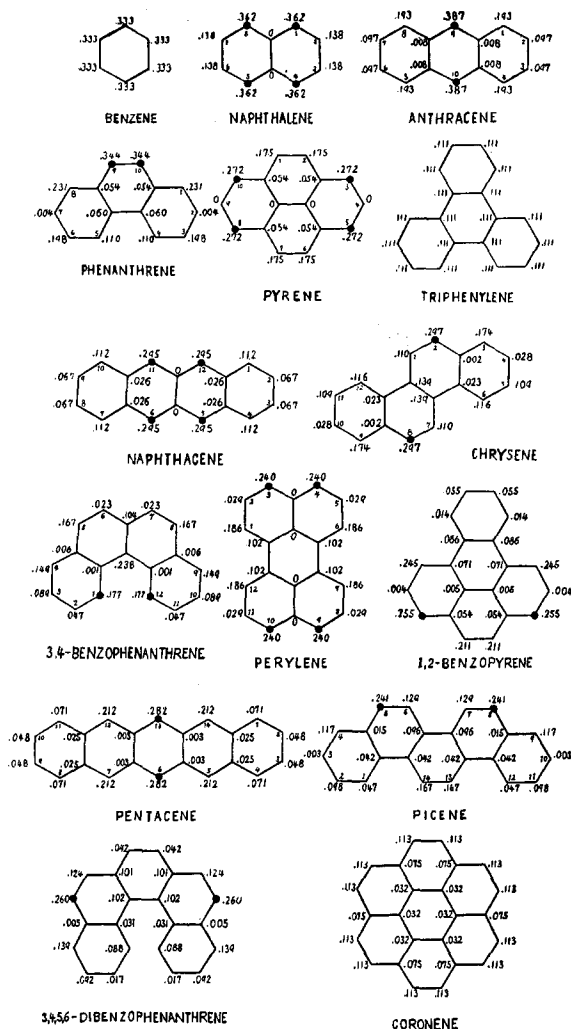


FIG. 1. The distribution of density of the frontier electrons.
 ●: Position of the largest frontier electron density.

In discussing the reactivity of the π -electron system in these molecules, we make here an assumption that the two electrons occupying the highest π -orbital in the ground state, distinct from the other π -electrons, are essential to the reactivity of the system. We shall refer hereafter to this sort of π -electrons as *frontier electrons*.

The discrimination of the pair of *frontier* π -electrons in relation to the reactivity is considered reasonable so far as, in photochemical processes in which a conjugated double bond system takes part, the triplet state has often been demonstrated as an intermediate step essential to the reaction.⁶

It is our opinion, furthermore, that for ordinary, nonphotochemical, organic reactions—not only for oxidation and other reactions of radical nature, but also, with certain reservations, for reactions of ionic character such as the electrophilic substitutions in aromatic

nuclei—a transition of the *frontier* electrons into the triplet state under the influence of the attacking reagent may be essential to the process of chemical activation. Thus, this will be described as a rate process in which the *frontier* electrons solely play an orienting or guiding role, so that the density distribution of the *frontier* electrons will determine the configuration of the transition complex for the reaction.

In view of these considerations we have numerically calculated the density distribution of the *frontier* electrons at each carbon atom in the molecule. The result for 15 hydrocarbons is shown in Fig. 1. The points marked in each diagram of molecules indicate the position of the largest density of the *frontier* electrons.

In this calculation the exchange integrals, except those between two adjacent carbon atoms, and all the overlap integrals, are neglected and the Coulomb integral, as well as the exchange integral, is assumed to be constant.

It is a simplifying circumstance for the present treatment that the distribution of the π -electron density in the triplet state is always the same as that for the ground state, so far as the hydrocarbon systems in the present consideration are concerned. We can, therefore, proceed with our discussion without settling the question as to which of these two states is essential to the reaction. This problem is discussed in some detail in the last section.

COMPARISON WITH EXPERIMENT

In Table I the structures of the reaction products of oxidation, halogenation, nitration, and sulfonation of ten condensed aromatic hydrocarbons are tabulated along with, in the second column, the positions of the highest density of the *frontier* electrons in each hydrocarbon indicated in Fig. 1. No experimental results could be found in the literature for 1,2-benzopyrene and 3,4,5,6-dibenzophenanthrene.

As can be seen from the Table I, there is a perfect agreement between calculation and the result of experiment in the cases of naphthalene, anthracene, phenanthrene, chrysene, naphthacene, perylene, pentacene, and picene.

In the case of 3,4-benzophenanthrene our calculation seems to disagree with the experiment. But this is explicable from the consideration that the position of the largest electron density, 1, is sterically hindered, so that the position 5 where the density is second to the position 1 may be attacked, in conformity with the experiment.

The apparent exceptions in the examples of sulfonation, that of anthracene in particular, may be attributed to the instability of primary sulfonation products, as is widely recognized by preparative organic chemists.

As to picenequinone, the 13,14-structure has been reported by Burg,⁷ but later on Cook⁸ proved experi-

⁶ G. N. Lewis and M. Kasha, J. Am. Chem. Soc. **67**, 994 (1945).

⁷ O. Burg, Ber. deut. chem. Ges. **13**, 1836 (1880).

⁸ J. W. Cook, J. Chem. Soc. **1941**, 685.

TABLE I. Comparison of the predicted positions of the highest reactivity with experimental results. "1-Cl" means, e.g., "1-Chloronaphthalene," "3,9-di-SO₃H" means, e.g., "Perylene-3,9-disulfonic acid," etc.

Compounds	Position of the highest density of the frontier electrons	Oxidation	Structure of the reaction products		
			Halogenation	Nitration	Sulfonation
Naphthalene	1,4,5,8	1,4-quinone ^a (5,8-quinone)	1-F ⁱ 1-Cl ^m 1-Br ⁿ 1,4-di-Cl ^o 1,5-di-Cl ^o	1-NO ₂ ^{ee} 1,5-di-NO ₂ ^{dd} 1,8-di-NO ₂ ^{dd}	1-SO ₃ H ⁱⁱ (2-SO ₃ H at higher temp.)
Anthracene	9,10	9,10-quinone ^b	9-Cl ^p 9,10-di-Cl ^q 9-Br ^p 9,10-di-Br ^r	9-NO ₂ ^{ee} 9,10-di-NO ₂ ^{ee}	(9-SO ₃ H unstable?) 1,5-, 1,8-di-SO ₃ H ⁱⁱ
Phenanthrene	9,10	9,10-quinone ^c	9,10-di-Cl ^s 9-Br ^t	9-NO ₂ ^{ff}	9-SO ₃ H ^{kk} (2 or 3-SO ₃ H at higher temp.)
Pyrene	3,5,8,10	3,8-quinone ^d 3,10-quinone ^e	3,5,8,10-tetra-Cl ^u 3,5,8-tri-Br ^v	...	3,5,8,10- tetra-SO ₃ H ^{ll}
Naphthacene	5,6,11,12	5,12-quinone ^f (6,11-quinone)	5,12-di-Cl ^w 5,12-di-Br ^w	...	...
3,4-Benzophenanthrene	1,12	5,6-quinone ^g	...	...	...
Chrysene	2,8	1,2-quinone ^h	2-Cl ^x 2-Br ^x	2,8-di-NO ₂ ^{gg}	2-SO ₃ H ^{mm}
Perylene	3,4,9,10	3,9-quinone ⁱ 3,10-quinone(?) ⁱ	3,9-di-Cl ^y 3,4,9,10-tetra-Cl ^z 3,9-di-Br ^{aa} 3,10-di-Br ^{aa}	3,10-di-NO ₂ ^{hh} 3,4,9,10-tetra-NO ₂ ^{hh}	3,9-di-SO ₃ H ⁿⁿ 3,10-di-SO ₃ H ⁿⁿ
Pentacene	6,13	6,13-quinone ^j	...	...	...
Picene	5,8	5,6-quinone ^k	13,14-di-Br(?) ^{bb}	...	...

- ^a C. E. Groves, *Ann. Chem.* **167**, 357 (1873).
^b J. Fritzsche, *J. prakt. Chem.* (1) **106**, 287 (1869).
^c R. Fittig and E. Ostermayer, *Ann. Chem.* **166**, 365 (1873).
^d R. Scholl and C. Seer, *Ann. Chem.* **394**, 125 (1912).
^e Rütgerswerke, A.-G. Germ. Pat. 613, 914 (May 31, 1935).
^f S. Gabriel and E. Leupold, *Ber. deut. chem. Ges.* **31**, 1277 (1898).
^g J. W. Cook, *J. Chem. Soc.*, 3524 (1931).
^h C. Liebermann, *Ann. Chem.* **158**, 309 (1871).
ⁱ A. Zinke and E. Unterkreuter, *Monatshfte Chemie* **40**, 408 (1919).
^j E. Phillipi, *Monatshfte Chemie* **53/54**, 639 (1929).
^k J. W. Cook, *J. Chem. Soc.*, 685 (1941).
^l Schiemann, Gueffroy, and Wikelmüller, *Ann. Chem.* **487**, 270 (1931).
^m A. Töhl and O. Eberhard, *Ber. deut. chem. Ges.* **26**, 2945 (1893).
ⁿ C. Glaser, *Ann. Chem.* **135**, 41 (1865).
^o J. S. Salkind and S. B. Faerman, *J. Russ. Phys.-Chem. Ges. (U.R.S.S.)* **62**, 1021 (1930).
^p W. H. Perkin, *Chem. News* **34**, 145 (1876).
^q Ed. Lippmann and I. Pollak, *Ber. deut. chem. Ges.* **34**, 2768 (1901).
^r C. Graebe and C. Liebermann, *Ber. deut. chem. Ges.* **1**, 186 (1868).
^s J. Schmidt and R. Schall, *Ber. deut. chem. Ges.* **39**, 3893 (1907).
^t L. F. Fieser and C. C. Price, *J. Am. Chem. Soc.* **58**, 1838 (1936).

- ^u I. G. Farbenind, A.-G. Fr., 261, 627 (March 23, 1934).
^v E. Clar, *Ber. deut. chem. Ges.* **60**, 1671 (1936).
^w I. J. Postovskii and R. G. Beyles, *Compt. rend. Acad. Sci. U.R.S.S.* **39**, 102 (1943).
^x I. G. Farbenind, A.-G. Fr., 793, 893 (February 3, 1936).
^y Zinke, Funke, and Pongratz, *Ber. deut. chem. Ges.* **58**, 330 (1925).
^z A. Zinke and K. Funke, *Ber. deut. chem. Ges.* **60**, 580 (1927).
^{aa} Zinke, Linner, and Wolfbauer, *Ber. deut. chem. Ges.* **58**, 328 (1925).
^{bb} O. Burg, *Ber. deut. chem. Ges.* **13**, 1837 (1880).
^{cc} R. Piria, *Ann. Chem.* **78**, 32 (1851).
^{dd} Ch. Gassman, *Ber. deut. chem. Ges.* **29**, 1243, 1521 (1896).
^{ee} J. Meisenheimer and E. Connerade, *Ann. Chem.* **330**, 164 (1903).
^{ff} J. Schmidt and M. Strobel, *Ber. deut. chem. Ges.* **36**, 2511 (1903).
^{gg} I. G. Farbenind, A.-G. Fr., 794, 534 (February 19, 1936).
^{hh} Zinke, Hirsch, and Brozek, *Monatshfte Chemie* **51**, 205 (1929).
ⁱⁱ O. N. Witt, *Ber. deut. chem. Ges.* **48**, 743 (1915).
^{jj} J. Houben, *Das Anthracen und die Anthrachinone* (1929), p. 109.
^{kk} A. Werner, *Ann. Chem.* **321**, 270 (1902).
^{ll} I. G. Farbenind, A.-G. Germ. Pat., 620, 635 (October 26, 1935).
^{mm} J. G. Farbenind, A.-G. Fr., 789, 461 (October 29, 1935).
ⁿⁿ C. Marschalk, *Chem. Zentr.* **I**, 1833 (1927).

mentally that the quinone obtained from picene by oxidation must have the 5,6-structure which coincides with our results.

In view of these excellent agreements of our theory with experiment we believe that some essential relations must exist between the reactivity and the density of the *frontier* electrons.

As for the significance of the absolute value of the *frontier* electron density, two examples may be of value to be mentioned here: triphenylene is more stable than naphthacene in relation to oxidation;⁹ and anthracene is more readily oxidized than phenanthrene.¹⁰ Compar-

ing the largest values of the density in those hydrocarbons having the same number of carbon atoms and π -electrons, we can observe a parallelism between the reactivity and the absolute value of the *frontier* electron density.

THEORETICAL DISCUSSION

In our treatments so far disclosed the physical ground of discriminating the highest occupied π -orbital in the ground state from the other π -orbitals is left untouched. We shall try here to add some theoretical considerations.

Anyhow, it can be concluded from the successful results obtained above that certain differences must exist between the *frontier* electrons and the other π -

⁹ Ward, Kirner, and Howard, *J. Am. Chem. Soc.* **67**, 246 (1945).

¹⁰ R. Anschütz and G. Schultz, *Ann. Chem.* **196**, 38 (1879).

electrons in respect to their behaviors in the course of electronic interaction during the chemical activation. This fact may be understood in two different ways: the one way of understanding consists in the possibility that the frontier electrons, on account of their valence-electron-like nature, can pass into the next higher orbital, forming a triplet state, presumably under the influence of the field arising from the approaching reagent. From the symmetry consideration of the wave function of the whole system, it seems to be quite reasonable that, in order to retain the singlet spin symmetry of the whole system, the spin state of two electrons in the C-H σ bond to be substituted by the electrophilic reagent is simultaneously changed into a triplet in the course of the reaction which is considered to take place adiabatically.

To understand that the triplet formation of the *frontier* electrons can be effective for bond formation between the molecule and the electrophilic reagent, we can make use of the following analogy. Considering that, as shown by Bartlett,¹¹ a strong bond of a $^3\Sigma_u^+$ state is formed between two $p\sigma$ orbitals, we may assume that between the hydrocarbon molecule and the electrophilic reagent in a transition complex a $\{(sp^3)\sigma, (sp^3)\sigma\}$ -like bond is formed in the same way by a participation of the *frontier* electrons with parallel spins.

The reagent is thus attracted to the position of the largest *frontier* electron density, while the hydrogen ion, formerly in the C-H σ -bond, may be repelled from the molecule because of the triplet property of the C-H σ -part of the electronic system in the transition complex.

On the other hand, the contributions of all the π -electrons in the molecule must, in general, be taken into account on the occasion of chemical reaction. From this point of view, we come to the second way of understanding. When it is assumed that in the electronic interaction with an electrophilic attacking reagent the highest occupied π -orbital would always act as *attractive* and the other π -orbitals as *repulsive*, it can be expected that the formation of a transition complex at the position of the largest *frontier* electron density might be preferred in consequence of the resultant effect of all π -electrons, in either attractive or repulsive orbitals, upon the potential energy of the whole system in the process of formation of the transition complex. In another expression, this way of understanding consists

in attributing the distinction of the highest occupied orbital from the others to the difference in the nature of each orbital in reference to the interaction with the attacking reagent. Of course, the repulsive effect represents a resultant contribution of all π -electrons other than *frontiers*. Some of them may give attractive effects but in this case the contribution must be smaller than those of *frontier* electrons. The possibility of a fact that a certain orbital in an atom attracts and that another orbital in the same atom repels a certain orbital of another atom has been shown by Araki and Watari¹² in the case of C₂-molecule.

The two ways of theoretical approach as discussed above may be considered as corresponding to the two different interpretations in the description of the process of electronic interaction. A correlation between them needs to be confirmed by further investigations.

The concept of the frontier electron, as well as the principle involved in the above discussion, is hoped to find applications more widely to problems on the reactivity of π -electron systems.

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The authors are grateful to Professor S. Kodama and Professor G. Araki for valuable suggestions and encouragements. Especially are the authors indebted to Professor Araki for enlightening them on the arguments developed in the part of discussion.

APPENDIX. SYMMETRY GROUP AND MODES OF REDUCTION OF SECULAR EQUATIONS.*

Compounds	Symmetry group	Modes of reduction of secular equations
Benzene	D_6	$\Gamma(\Psi) = \Gamma A_1 + \Gamma B_1 + \Gamma E_1 + \Gamma E_2$
Coronene	D_6	$\Gamma(\Psi) = 3 \cdot \Gamma A_1 + \Gamma A_2 + \Gamma B_1 + 3 \cdot \Gamma B_2 + 4 \cdot \Gamma E_1 + 4 \cdot \Gamma E_2$
Triphenylene	D_2	$\Gamma(\Psi) = 3 \cdot \Gamma A_1 + 3 \cdot \Gamma A_2 + 6 \cdot \Gamma E$
Anthracene	D_2	$\Gamma(\Psi) = 4 \cdot \Gamma A_1 + 3 \cdot \Gamma B_1 + 3 \cdot \Gamma B_2 + 4 \cdot \Gamma B_3$
Naphthalene	D_2	$\Gamma(\Psi) = 5 \cdot \Gamma A_1 + 4 \cdot \Gamma B_1 + 4 \cdot \Gamma B_2 + 5 \cdot \Gamma B_3$
Naphthalene	D_2	$\Gamma(\Psi) = 3 \cdot \Gamma A_1 + 2 \cdot \Gamma B_1 + 2 \cdot \Gamma B_2 + 3 \cdot \Gamma B_3$
Pentacene	D_2	$\Gamma(\Psi) = 6 \cdot \Gamma A_1 + 5 \cdot \Gamma B_1 + 5 \cdot \Gamma B_2 + 6 \cdot \Gamma B_3$
Perylene	D_2	$\Gamma(\Psi) = 6 \cdot \Gamma A_1 + 4 \cdot \Gamma B_1 + 6 \cdot \Gamma B_2 + 4 \cdot \Gamma B_3$
Pyrene	D_2	$\Gamma(\Psi) = 5 \cdot \Gamma A_1 + 3 \cdot \Gamma B_1 + 3 \cdot \Gamma B_2 + 5 \cdot \Gamma B_3$
3,4-Benzo-phenanthrene	C_2	$\Gamma(\Psi) = 10 \cdot \Gamma A + 8 \cdot \Gamma B$
1,2-Benzopyrene	C_2	$\Gamma(\Psi) = 10 \cdot \Gamma A + 10 \cdot \Gamma B$
3,4,5,6-Dibenzo-phenanthrene	C_2	$\Gamma(\Psi) = 11 \cdot \Gamma A + 11 \cdot \Gamma B$
Phenanthrene	C_2	$\Gamma(\Psi) = 7 \cdot \Gamma A + 7 \cdot \Gamma B$
Picene	C_2	$\Gamma(\Psi) = 11 \cdot \Gamma A + 11 \cdot \Gamma B$
Chrysene	C_2	$\Gamma(\Psi) = 9 \cdot \Gamma A + 9 \cdot \Gamma B$

* $\Gamma(\Psi)$: reducible representation. $\Gamma A, \Gamma B, \dots$: irreducible representations of character $A, B, \dots$.

¹¹ J. H. Bartlett, Phys. Rev. **37**, 505 (1931).

¹² G. Araki and W. Watari, Prog. Theoret. Phys. **6**, 945 (1951).